“On the way with Python”

Lecture

02

Python Variables

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What is Variable?

- A variables are containers for storing data values. The value of a variable can be changed during program execution.

```
name = “Rohim”       # Assigning a String value
age = 25             # Assigning an Integer value
grade = 89.3     # Assigning a Float value
is_married = True # Assigning a Boolean value
```

Why use Variable?

Variables are used to:

- Store data
- Reuse values
- Make programs readable and manageable
- Perform calculations easily

Variable declaration in Python

Python automatically determines the data type based on the value.

```
name = “Rohim”
age = 25
```

Rule of Naming Variables

- An identifier can only contain alphanumeric characters (a-z, A-Z, 0-9) and underscores (_). It cannot start with a digit.
- Python is case-sensitive, so myVar and myvar are different identifiers.
- Identifiers cannot be a reserved keyword.
- There is no limit on the length of an identifier, but it's recommended to keep it concise and meaningful.

Valid Variables

valiable.py

```python
myvar = "John"
my_var = "John"
_my_var = "John"
myVar = "John"
MYVAR = "John"
myvar2 = "John"
```

Invalid Variables

```
1age   # starts with number
user-name  # hyphen not allowed
class  # keyword
```

Multi Word Variable Names

Camel Case:

- Each word, except the first, starts with a capital letter:

CamelCase.py

```
myVariableName = "John"
```

Pascal Case:

- Each word starts with a capital letter:

pascalcase.py

```
MyVariableName = "John"
```

Snake Case

- Each word is separated by an underscore character:

snakcase.py

```
my_variable_name = "John"
```

Many Value to Multiple **Variables**

Note: Make sure the number of variables matches the number of values, or else you will get an error.

variable.py

```
x, y, z = "Orange", "Banana", "Cherry"
print(x)
print(y)
print(z)
```

One Value to Multiple Variables

variable.py

```python
x = y = z = "Orange"
print(x)
print(y)
print(z)
```

Checking Variable Type

```
x = 100
print(type(x))
```

Variable Type Casting

- Converting one data type to another

```python
x = "10"
y = int(x)
print(y)
```

Local Variables

- Defined variable inside a function

```python
def show():
    x = 10
    print(x)
```

Global Variables

```python
x = 20
def show():
    print(x)
```

Global Variables

- Global variables can be used by everyone, both inside of functions and outside.
- Variables that are created outside of a function are known as global variables.

```
x = 20
def show():
    print(x)
```

Global Variables by Global Keyword

- If you use the global keyword, the variable belongs to the global scope:

global_variable.py

```python
def myfunc():
  global x
  x = "fantastic"

myfunc()

print("Python is " + x)
```

Deleting a Variable

```python
x = 10
del x
```

Common Errors with Variables

- Using undefined variable

```
print(x)  # NameError
```

- Wrong type usage

```
x = "10"
print(x + 5)  # TypeError
```

“Thanks All”